



POLITECNICO
MILANO 1863

High Performance Scientific Computing for Aerospace Project

Progress Update: Solvers & Parallelization

December 14th, 2025

Current Achievements

- **Monodimensional Solver:** We have successfully validated the 1D solver logic; it is now functional.
DICIAMO DELLA PRESSIONE
NON IMPLEMENTATA?
- **SIMD Implementation:** The *Solver Class* has been optimized using OpenMP directives to leverage data-level parallelism.

Figure 1: Execution time comparison (Scalar vs. Vectorized).

Algorithmic Compatibility

- Our current *Solver* implementation inherently supports the Schur complement method without requiring major refactoring.
- The class structure allows for modular handling of sub-domains. we will have to implement a **Rank-Based Logic**

